

Deciding What to Do With a Review You Cannot Verify: A Cost-Sensitive Belief Agent Using Text and Star Rating Only

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Abstract

An e-commerce moderation agent observes a newly submitted review’s text and star rating and must permit, flag, route to a human, or hide it, without knowing whether the review is a genuine customer experience, a paid or machine-generated fake, or a non-independent review written by someone close to the business. We build a small agent that maintains an explicit belief over those three hidden states — though the third carries zero prior mass in the main experiment, which is therefore a two-state problem — and selects an action by minimising expected cost under an asymmetric, volume-weighted cost matrix. We compare it against a keyword baseline and against a fixed-threshold version of itself on a 40-case held-out set and a 12-case adversarial probe set. The agent improves precision from 0.647 to 0.933 and recall from 0.550 to 0.700 on the like-for-like accounting (0.778 if the cases it declines to decide are dropped from the denominator), but the result does not survive scrutiny intact: the baseline beats the agent on the probe set (recall 0.857 versus 0.333), the agent is poorly calibrated (ECE 0.145), an ablation shows the six hand-designed features contribute a single case beyond the lexical channel, and the ranking of the two policies reverses depending on whether routed cases are counted. Six design decisions came out of four public discussion threads with practitioners and changed the agent in ways the measurements did not, including the addition of the third hidden state. We report the negative findings in full because they are the informative part.

1 Introduction

A platform receives a review. It has, at that instant, two things: some text and a number of stars. It does not have the reviewer’s purchase history, IP address, account age, or photo metadata — either because the reviewer is new, or because the platform is small, or because the decision has to be made before those signals are joined. It must nonetheless do something with the review.

Stated in the form the assignment requires:

The agent observes a newly submitted customer review text and star rating. It must select permit, warn, hide, or report because whether the post is a genuine customer experience versus a paid spam or competitor fake review is not known.

The constraint — text and rating, nothing else — is deliberate and is enforced in code: the agent’s entry point takes `(text, rating)` and no other argument. It is also the constraint a practitioner immediately objected to. In [r/LanguageTechnology, u/Mundane_Ad8936](#) stated that a single review cannot be classified without user history or graph features. We agree, and the project is scoped accordingly: this is a cold-start triage filter whose most important output is often the admission that it does not know.

This preprint’s contribution is not a detector. It is a decision procedure under a belief, plus an honest account of where that procedure fails. The findings we consider most useful are negative:

1. The agent beats a keyword baseline on held-out data and loses to it on a probe set designed to attack its assumptions.
2. Its beliefs are not calibrated (ECE 0.145), and 60% of them are pinned by a clipping constant rather than by evidence.
3. The six interpretable features that motivate the whole design contribute one case out of forty beyond a plain unigram channel.
4. Which of the two policies is “better” depends entirely on whether the cases it declined to decide are counted in the denominator.

2 Related Work and User Discussions

Prior work. Ott et al. [2011] established that deceptive opinion spam is separable from genuine review text by psycholinguistic and n -gram features, and produced the first gold-standard deceptive review corpus. Their positive class is crowdworker-authored deception. Salminen et al. [2022] produced the corpus used here, in which the positive class is GPT-2-generated review text paired with genuine Amazon reviews drawn from the corpus of Ni et al. [2019]. Elkan [2001] gives the decision-theoretic result this agent’s policy rests on:

with a known cost matrix, the optimal action is the one minimising expected cost under the posterior, and thresholds on probability are a special case rather than the general rule. Guo et al. [2017] is the source of the expected-calibration-error estimator we report and of the observation that a confident classifier is not thereby a calibrated one. Sap et al. [2019] document that lexical classifiers in content moderation systematically penalise African-American English; we cite it because it names an untested risk in this system, not because we tested for it.

Discussions with people. Six design decisions in this agent came from four public discussion threads rather than from the literature or from measurement (`discussion-record.md` records ten changes in total; these are the six that reach the code). Each is recorded with a link in `discussion-record.md`.

- **Probability bands.** Practitioners in `r/trustandsafetypros` described “warn” as soft friction or a queue router rather than a public label, and put autonomous removal behind high confidence. The policy was rewritten as explicit bands: permit below 0.50, human queue 0.50–0.84, act at 0.85 and above.
- **Flag instead of hide.** A seller in `r/Amazonsellercentral` objected to autonomous hiding and preferred a flag carrying an explanation for manual approval. Policy A’s high-confidence action was changed accordingly.
- **Volume-weighted costs.** A commenter in `r/Yelp` stated that the damage from a wrongly removed review depends on how many reviews the business already has. Every cost of acting wrongly against a genuine review, or against a `solicited` one, is now multiplied by $v = \min(1, 200/\text{total_reviews})$.
- **Specificity over sentiment.** Two `r/Yelp` users independently said a genuinely angry customer names falsifiable detail — a dish, a quantity, a sequence — while a planted review stays vague. The agent’s strongest hand-designed feature keys on this, and star/text mismatch was demoted to a secondary signal.
- **Specificity qualified.** The same user then said fakers lift dish and staff names from the reviews already on the page. Specificity separates lazy fakes from real reviews and fails against motivated ones. It is reported as non-robust rather than as a solution.
- **A third hidden state.** `u/ADrPepperGuy` clarified that by “fake” he meant a friend or family member writing a review for someone they know. The visit happened and the detail is true, so the specificity signal is blind to it by construction. The state `solicited` was added for this reason.

3 Agent Design

Input. A review’s text and its star rating. Nothing else.

Hidden states. $\mathcal{S} = \{\text{genuine}, \text{fake}, \text{solicited}\}$: an independent customer; a paid, competitor, or machine-generated review; and a review by someone with a personal

relationship to the business, whose visit was real but whose review is not independent.

Belief. A full posterior over $\mathcal{S}$, not a score.

Actions. $\mathcal{A} = \{\text{permit}, \text{flag}, \text{route}, \text{hide}\}$, written in full as `flag_with_explanation` and `route_to_human` in the code. These realise the assignment’s permit/warn/hide/report: `flag` is the warn action and `route` is the report action, both named after the form the `r/trustandsafetypros` practitioners described.

Human reasoning function. One only: identify uncertainty and hand a high-cost decision to a person rather than guessing. Both policies may decline to decide. No other human faculty is imitated.

Evidence channels. Six discrete interpretable features are extracted from text and rating: `specificity` (3 levels), `generic_praise` (3), `length_band` (4), `polarity_mismatch` (2), `experience` (2), `exclamation` (3). Separately, a unigram Naive Bayes channel is fit on a 6,000-row training split (vocabulary 3,835 terms after a minimum document frequency of 5), producing a log-likelihood ratio that is clipped at ± 4.0 .

Feedback. `FeedbackLog` records human verdicts for cases sent to the queue and for enforcement actions that are later appealed. Permitted reviews generate nothing, which is a structural limitation identified in `r/learnmachinelearning` and discussed in Section 8.

4 Probability Model and Decision Rule

4.1 Belief update

Let $f_1, \dots, f_6$ be the observed feature levels and $w_1, \dots, w_n$ the review’s in-vocabulary tokens. Treating the channels as conditionally independent given the state,

$$P(s \mid f, w) \propto P(s) \prod_{i=1}^6 P(f_i \mid s) \cdot \exp(\lambda_s(w)), \quad (1)$$

computed in log space. The feature likelihoods $P(f_i \mid s)$ are estimated on the training split with Laplace smoothing:

$$P(f_i = \ell \mid s) = \frac{c_s(f_i = \ell) + 1}{c_s + L_i}, \quad (2)$$

where L_i is the number of levels of feature i . This holds for genuine and fake only. The `solicited` row is not fitted: five of its six features are copied from `genuine` and `generic_praise` is reweighted toward heavier praise by a stipulated factor (`SOLICITED_ASSUMPTION` in the code). The lexical term is the clipped log-likelihood ratio

$$\lambda_s(w) = \text{clip}(\Lambda(w), -\tau, \tau), \quad \tau = 4.0, \quad (3)$$

$$\Lambda(w) = \sum_{j=1}^n \log \frac{P(w_j \mid \text{fake})}{P(w_j \mid \text{genuine})},$$

added with sign $+$ for fake and set to exactly 0 for both `genuine` (the reference state) and `solicited`. The lexical channel therefore contributes Λ , not 2Λ , to the fake-versus-genuine log-odds, and contributes nothing at all to

solicited — an admission of ignorance whose consequence we quantify in Section 8.

The quantity both policies act on is $p = P(\text{fake}) + P(\text{solicited})$.

4.2 Cost matrix

Costs are in relative units. With volume factor $v = \min(1, 200/\text{total_reviews})$:

Action	genuine	fake	solicited
permit	0	5	2
route	1	1	1
flag	$2v$	1	$1.5v$
hide	$10v$	0	$6v$

Only the presence of v is evidence-based, from the r/Yelp conversation. The magnitudes are stipulated; no seller answered the direct question about what a wrongly removed review costs, across two subreddits and one X post.

4.3 The two policies

Policy A (fixed bands) acts on p alone: permit if $p < 0.50$, route_to_human if $0.50 \leq p < 0.85$, and flag_with_explanation if $p \geq 0.85$. It is never permitted to hide.

Policy B (expected cost) follows Elkan [2001] and selects

$$a^* = \arg \min_{a \in \mathcal{A}} \sum_{s \in \mathcal{S}} P(s | f, w) C(a, s), \quad (4)$$

with ties broken toward the least destructive action (permit < route < flag < hide) and the tie noted in the decision reason. Policy B may hide.

Comparing A with B was intended as a direct test of the design change made after the r/AmazonSellerCentral discussion. It is not one. Under this cost matrix flag is weakly dominated by route for every belief, so Policy B never flags — zero flags on both datasets — and its only enforcement action is hiding. The comparison confounds two changes: threshold-versus-cost, and flag-only-versus-hide-only.

5 Test Method

Data. The Fake Reviews Dataset of Salminen et al. [2022], 40,432 rows labelled CG (computer-generated) and OR (original human). Splitting is deterministic under seed 20260817 into a 6,000-row fit split and a 40-case held-out test set balanced 20/20.

Labelling provenance. This is stated explicitly because a r/learnmachinelearning commenter asked for it. The test-set labels are the dataset’s own CG/OR labels, decided by the dataset authors’ construction procedure, not by us. The 12-case probe set is hand-written by the author, and its labels are the author’s by construction — which is why it is reported separately and never pooled with the 40.

Probe set. Twelve cases written to attack specific assumptions: solicited reviews with true detail, fakes with detail harvested from a product page, sarcasm with inverted polarity, and terse genuine reviews. It runs under a different prior (0.60/0.30/0.10) because solicited has zero prior mass in the main run and cannot otherwise be inferred; probe and test numbers are therefore not directly comparable.

System	Prec.	Rec.	Rec. [†]	Cost	ECE
baseline_keyword	0.647	0.550	0.550	68.0	n/a
fixed_band (A)	0.933	0.778	0.700	42.0	0.145
expected_cost (B)	0.929	0.867	0.650	31.0	0.145

Table 1: Held-out test set, 40 cases (20 genuine / 20 fake). † counts routed cases in the denominator. Human-review rate: 0%, 15.0%, 27.5%. The cost column is reported but is not a valid comparison — see the text.

Baseline. A flat keyword rule with no belief, no cost model, and no option to defer. Its term-count threshold is tuned on the fit split to maximise F_1 (tuned value 1). Its keyword list was generated with LLM assistance, following a suggestion in r/learnmachinelearning. It cannot settle the follow-up question we asked there — whether LLM-generated rules inflate false positives — because there is no hand-written control list to compare against; the lexical-only ablation row serves as the stronger comparator instead.

Metrics. Precision and recall against the positive class “not genuine”; expected calibration error over five equal-width bins following Guo et al. [2017]; total cost under the cost matrix; and the human-review rate. Two recall figures are reported throughout, excluding and including routed cases in the denominator, for reasons given in Section 6.

Reproduction. Three seeded commands, Python 3.11, no third-party packages (the figure script uses matplotlib), no network. The seventeen parameters in results/run_manifest.json cover the splits, priors, thresholds and lexical settings; the two smoothing constants, the solicited reweighting factors and the cost magnitudes live in the source and are not in the manifest. The source dataset is not checksummed, so the numbers reproduce exactly against the same download and are not guaranteed against a re-download. Code: <https://github.com/kishore461/e-commerce-review-moderator-agent>.

6 Results

Both agent policies beat the baseline on precision and recall on the held-out set. Four qualifications apply, and they are the substance of this section.

The cost comparison is circular. Policy B selects the action minimising expected cost under a cost model and is then scored with that same cost model. “B has the lowest cost” is a property of Equation (4), not a finding. The column is printed for completeness and excluded from every claim.

The policy ranking reverses. Excluding routed cases from the denominator flatters a policy that declines — an agent routing 39 of 40 cases and getting the last one right would score perfectly. Including them penalises a policy for correctly identifying its own uncertainty, and the baseline cannot route at all, so only that version is like-for-like. Under the first, B leads (0.867 vs. 0.778); under the second, A leads (0.700 vs. 0.650). We report both and claim neither policy is better.

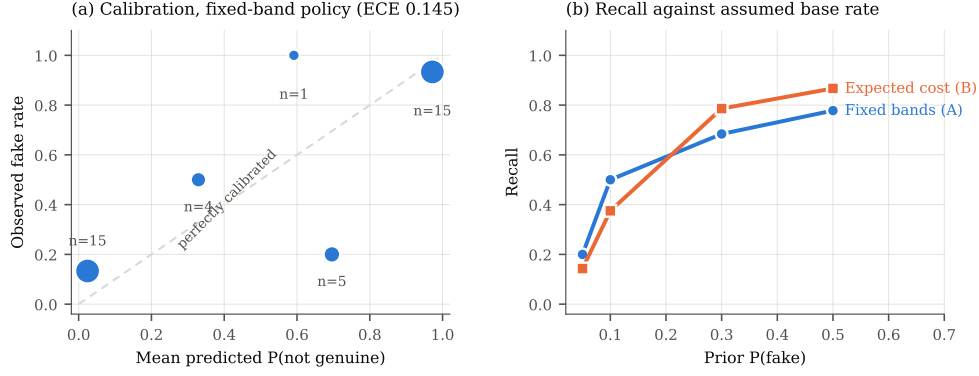


Figure 1: (a) Reliability of the agent’s beliefs on the held-out set; marker area is proportional to bin count. The $[0.6, 0.8]$ bin sits far below the diagonal but holds five cases. (b) Recall against the assumed base rate. Both policies degrade sharply as $P(\text{fake})$ falls toward a realistic deployment value.

Channels	Pol.	TP	FP	FN	Rt.	Prec. / Rec.
feat.+lex.	A	14	1	4	6	0.933 / 0.778
feat.+lex.	B	13	1	2	11	0.929 / 0.867
lex. only	A	13	2	4	6	0.867 / 0.765
lex. only	B	13	1	2	10	0.929 / 0.867
feat. only	A	0	0	5	22	— / 0.000
feat. only	B	0	0	0	40	— / —

Table 2: Channel ablation on the held-out set. “Features only” under Policy B routes every case, so it has no decisions to score.

The baseline wins on the probe set. On the 12 hand-written adversarial cases the baseline reaches recall 0.857 against 0.333 for both agent policies. The probe set was built to attack the agent’s assumptions and it succeeds.

The features contribute almost nothing. Table 2 removes each channel. With the feature terms genuinely omitted from Equation (1) — not set to level zero, which injects -1.86 log-odds and is not removal — the lexical channel alone reaches recall 0.765 (A) and 0.867 (B) against 0.778 and 0.867 for the full system: one true positive on A, none on B. On A they also remove one false positive (precision 0.933 against 0.867), which is the larger of their two contributions and is invisible in a recall-only table. Alone they never reach the acting threshold at all: under Policy A they permit 18 cases and route the other 22, giving recall 0.000; under Policy B they route all 40. An earlier version of this work reported that features raised recall from 0.632 to 0.778; that was an artefact of the level-zero bug, found in review and corrected.

The headline numbers do not travel. Table 1 assumes a 50% base rate of fakes, which no deployment has. Sweeping the prior (Figure 1b) gives recall $0.778 \rightarrow 0.500 \rightarrow 0.200$ for Policy A and $0.867 \rightarrow 0.375 \rightarrow 0.143$ for Policy B as $P(\text{fake})$ falls from 0.5 to 0.1 to 0.05.

Calibration. ECE is 0.145 on the test set and 0.211 on the probe set. The $[0.6, 0.8]$ bin predicts a mean of 0.696 against an observed rate of 0.200, but holds five cases; bin counts are printed beside every reliability table for that reason.

7 Failure Analysis

Routed cases are not counted as failures — declining to decide is a permitted action. `results/failures.md` names nine failure conditions across the two datasets; the five largest are described here, and they cover 29 of the 35 incorrect decisions. The counts in parentheses are counts of *incorrect decisions*, pooled across all three systems, not of distinct cases — the same review can be got wrong by more than one system.

SHORT-FAKE-MISSED (9 decisions, 5 cases). Too little text to separate from a short genuine review. “*Came broken and cracked frame. The only way to fix it was to just fix the frame*” — true state `fake`, 1 star, permitted by both the baseline and Policy A at $p = 0.339$.

SPECIFIC-GENUINE-FLAGGED (8 decisions, 7 cases). A detailed real review acted on anyway. This produces the single most expensive decision in the experiment: “*My dog loves this toy and is always chewing on it and he has nice strong, clean teeth. This is my 3rd one*” — genuine, 4 stars, and Policy B assigns $p = 0.983$ and *hides* it, at a cost of 10.0 units, the maximum any single decision can incur. Policy A flags the same case. This is the case that justifies the `r/Amazonsellercentral` objection to autonomous hiding.

FLUENT-FAKE (5 decisions, 3 cases). Long, fluent machine text with no checkable content, permitted at $p = 0.019$ — the agent is confidently wrong, not uncertain.

SOLICITED-INVISIBLE (5 decisions, 2 probe cases). Both agent policies permit both solicited probe cases; only the keyword baseline flags one of them, by accident. “*My sister runs the front of house here and I eat here most Fridays*” — an explicit statement of non-independence, and the agent scores it $p = 0.101$. The feature that was supposed to catch fakes (specificity) is exactly the feature that certifies this review as genuine, because its specifics are true.

SPECIFICITY-HARVESTED (2 decisions, 1 probe case). Detail lifted from the product page, permitted at $p = 0.294$ — the failure `u/XxLogitech98xX` predicted before we tested it. A later public discussion qualified it in the agent’s favour.

An engineer who had used a model fine-tuned to evade AI detection reported that its output was unusable prose: “*even if it manages to evade bot detection, it’s useless*”. If that generalises, a moderation signal does not have to be unbeatable — only expensive enough to beat that the resulting fake is not worth posting. We have not measured that cost, and it is the difference between a defeated signal and an expensive one.

Pooled across all three systems and both datasets — so this is not a per-system cost — permitting a fake occurs 19 times at 5.0 units (95.0 total), flagging a genuine review 10 times at 2.0 (20.0), and hiding a genuine review once at 10.0. The rare catastrophic error and the common cheap one carry comparable weight, which is what the cost model exists to express.

8 Limitations, Ethics, and Human Control

The README records nineteen limitations. The six that most constrain the conclusions:

1. **The dataset’s labels are not the project’s hidden states.** CG/OR is machine-versus-human authorship. The problem statement concerns a *paid human* writing a fake. The experiment tests one corner of *fake* and says nothing about the case that motivated the work.
2. **Leakage lands on the positive class.** At token-Jaccard > 0.30 , 11 of 20 test fakes have a near-neighbour in the fit split against 0 of 20 genuine, because machine-generated rows repeat near-verbatim spans. This inflates recall specifically.
3. **solicited is structurally undetectable.** Its likelihoods are copied from genuine for five of six features and its lexical LLR is fixed at zero, so $P(\text{solicited})/P(\text{genuine})$ is bounded above by 0.382 for any text at all — the highest ratio observed on the probe set is 0.38218. It is not inferred from the text; it is its prior times one of three constants, selected by the `generic.praise` level alone.
4. **Most beliefs are set by a constant.** Raw LLRs span roughly -79 to $+82$ and 24 of 40 test cases saturate the ± 4.0 clip exactly. For those, the belief is a step function of the LLR’s sign, τ is the most influential parameter in the system, and the 0.85 threshold is barely load-bearing — 0.80 and 0.85 produce identical decisions here.
5. **Evidence is double-counted.** Equation (1) multiplies the feature and lexical channels as if conditionally independent, but the unigram model contains the very words the features count. This is the likely source of the overconfidence in (4) and it is not fixed.
6. **No dialect or demographic bias audit has been run.** Unigram filters in moderation are documented to penalise non-standard dialects and ESL syntax [Sap *et al.*, 2019]. Two features make this worse rather than better: `length.band` treats brevity as evidence and `specificity` rewards one register of concrete detail. A writer with limited English leaving a short, plain review sits exactly where this model is most confident. This is the most consequential untested risk in the project.

Ethics and human control. The customer who wrote the review is not in the cost model. They lose their words with no notification and no appeal, and the volume factor scales that harm by the *seller’s* review count — the person harmed is not the person the cost is measured against. Human review is modelled as free, instant, and infallible at a flat cost of 1.0, and the experiment feeds ground truth in as the human verdict, so the printed overturn rate is a relabelled ground-truth count and not feedback. Sweeping the volume parameter shows the human-control mechanism switching itself off: at 20,000 reviews Policy B routes nothing at all and hides 27 of 40 cases, because hiding becomes cheaper than a person’s time. The stated human reasoning function disappears exactly for the largest sellers, which are the deployments where it would matter most. Nothing here should be connected to a live enforcement path.

9 Conclusion and New Questions

A belief plus a cost matrix produces better decisions than a keyword rule on matched data, and worse decisions on data built to attack it. The parts of the design that came from talking to practitioners — the deferral band, flagging instead of hiding, scaling costs by review volume, and the third hidden state — changed the agent more than any measurement did. Three of the four are exercised somewhere in the experiment (the deferral band by the human-review rate, volume scaling by the cost sweep, the third state by the probe set), but none is validated against the outcome the practitioners actually cared about: what happens to a real seller and a real customer.

Open questions we cannot answer from here:

- What does a wrongly removed review actually cost a seller? Asked across two subreddits and one X post; no seller answered. Every cost magnitude in this paper rests on that gap.
- Does a one-sided feedback signal converge somewhere wrong? Permitted reviews generate no correction, so live measurement can only ever see false positives.
- Can `solicited` be given evidence rather than a prior? As built it is an assumption wearing the costume of a state.
- How much of the reported performance survives a dialect audit? On the argument in Section 8, item 6, we expect the answer to be uncomfortable.
- When a rewrite evades a detector, does it lose the author’s voice or the passage’s meaning? Those cost an attacker very different amounts to repair, and the difference decides how much protection the evasion-is-expensive argument actually buys.

AI-Use Statement

This project was built with AI assistance (Anthropic Claude, with independent review passes by Google Gemini). AI wrote most of the Python and drafted this document. AI did not participate in any public discussion: every Reddit and X contribution recorded in `discussion-record.md` was written

and posted by the author in the author’s own words, as the assignment requires. The design decisions listed in Section 2 came from those human conversations, not from a model. Seven AI review passes producing 56 comments are logged in `review-record.md` with an accept-or-reject decision and a reason for each; the ablation bug in Section 6 and the arithmetic-presentation error in the decision record were both found that way. Cases where an AI assistant asserted something false — including a set of fabricated X accounts — are recorded as AI errors in `research-file.md`. Every number in this preprint was checked against the committed outputs in `results/`, or recomputed from source, in a dedicated audit pass run after the first draft compiled. That pass found nine numeric or wording errors in the draft — including a stale bound of 0.402 where the current likelihood table gives 0.382 — and they are corrected here.

Contribution Statement

This is single-author work. Problem selection, public discussions, agent design, software development, tests, citation checks, preprint sections, and PDF checks were all carried out by the author, with AI assistance as described above.

Status of This Preprint

This is a course preprint written in the IJCAI-ECAI 26 style because the assignment requires that style. It has not been submitted to IJCAI, and IJCAI has not accepted it.

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